

Design and Validation of Configurable Online Aging Sensors in Nanometer-Scale FPGAs

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Abstract—In current CMOS nanometer technologies, aging effects may appear after relatively short operating times, compared to the expected lifetime of circuits. Therefore, there is an increasing need for on-chip aging monitoring, especially in high-performance, safety critical systems. This paper presents a programmable aging sensor that can be embedded in field-programmable gate array (FPGA)-based designs, using standard resources available in those devices and with minimal impact on the standard FPGA design flow. Given the limited amount of resources required by the sensor, it can be instantiated not only in the critical paths of a circuit, but also in those that may be identified to be more likely affected by aging effects. Experimental results, obtained in circuits of increasing complexity (where several sensors need to be used), are presented and discussed, demonstrating the good performance of the proposed sensor, as well as its low cost in terms of area overhead and power consumption. Results of aging experiments based on the standard US MIL-STD-883 Method 1015.50 “Burn-In Test” are also reported, demonstrating that the effect of aging on the sensor is negligible compared to that on the circuit under test, which is a key point for practical applicability. The proposed approach provides a novel and efficient solution to the specific FPGA design problems in this context, which are different from those addressed in application-specific integrated circuit design.

Index Terms—Aging sensor, delay faults, field-programmable gate array (FPGA).

I. INTRODUCTION

DIGITAL circuit operation using CMOS nanometer technologies faces increasing functional uncertainty, as device

scaling down brings new design and test challenges and new physical defects. Reducing MOSFET gate length causes high variability in new technology nodes, related to the increased dependence of path delay on many variables (like process, power-supply voltage, or temperature variations—PVT) [1]. Moreover, the impact of parametric variations on circuit correct functionality is a hot topic, not only in terms of “classic” (PVT) variations, but also of aging-induced variations, when extended long operation times (system operating for 5–10 years, e.g., in automotive, medical, and space applications) are to be considered. Aging effects caused by phenomena like negative bias temperature instability (NBTI), hot carrier injection (HCI), or time-dependent dielectric breakdown, among others, are gaining increasing relevance in new technologies. For emerging nanoscale technologies (namely FinFET technologies—20, 14, 10 nm), current research is starting to address the problems of variability, reliability, and aging (due to NBTI and HCI effects) [2]–[4]. The effects of all the signal integrity problems that affect a circuit (operation dependent or not) may lead to chip failures and delayed schedules to put a new product on the market, especially if cumulative effects are considered [5], [6]. Increased susceptibility and parameter variations in nanometer technologies have shown that ten years operation could increase delays in 20% [7]. However, this value varies with the technology used and with the operating conditions [8].

The most important long-term effect degrading circuit performance has been identified as the NBTI effect [9]. It affects primarily PMOS transistors, increasing their threshold voltage $|V_{thp}|$ over time. This will ultimately cause delay faults, as delayed responses will lead to erroneous capture of logic values in memory cells. The impact of NBTI on digital circuit performance has been under research and NBTI modeling in static and dynamic operation has been also proposed [10]. Solutions to make circuits more tolerant to V_{thp} degradation and to restrict NBTI performance degradation have been presented in [9], [11], and [12]. More general solutions to increase circuit’s dependability in the presence of aging effects were also presented in [13]–[15], with various aging sensor topologies aimed at globally detecting circuit’s performance degradation. In fact, built-in aging sensors are being under investigation for application-specific integrated circuits (ASICs) typically monitoring locally abnormal delays in signal paths with long propagation delay times [16], [17].

Although the impact of variability on system performance has been extensively studied in the context of ASIC-based design, limited research has been carried out for field-programmable

Manuscript received September 28, 2012; accepted February 28, 2013. Date of publication March 21, 2013; date of current version July 3, 2013. This work was supported in part by the Spanish Ministerio de Economía y Competitividad through Project TEC2010-21429-C02-01, including European FEDER funds.

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Digital Object Identifier 10.1109/TNANO.2013.2253795

gate array (FPGA) devices, which are also fabricated using state-of-the-art nanometer-scale IC technologies (e.g., 28 nm technology for Virtex7 Xilinx devices).

There is an increasing use of reconfigurable circuits (namely FPGAs) to replace ASICs in low-volume applications. All FPGA resources (memory, logic, interconnections) are sensitive to aging, and this phenomenon can become relevant in high-performance applications, especially those with high mission profiles (long uptimes and extended lifetime). Moreover, high-performance safety-critical applications require high dependability, which can justify the implementation of aging sensor functionality on these devices [16]–[19]. Hence, it is rewarding to adapt traditional hardware solutions (or to propose novel solutions) to monitor circuit aging in FPGA designs [18].

The operation of built-in aging sensors is based on detecting and flagging excessive delays in the critical paths (CPs) of a circuit, that is, in signal paths with long propagation delay times. The main difference between monitoring aging effects in an ASIC or in an FPGA is that in the case of ASICs, the CPs are well defined. As ASICs are designed for a specific function, the CPs of the circuit can be identified during the design process. Therefore, built-in aging sensors can be included in the ASIC layout and implemented at CMOS transistors level. For FPGAs, this is not possible because CPs in FPGAs are application dependent. FPGAs consist of a set of resources (general-purpose logic, interconnections, memory blocks, clock management blocks, etc.) that can be configured to support a user-defined application. Consequently, each different application implemented in an FPGA has different CPs. Thus, the placement of the aging sensors depends on the application. Sensors cannot be implemented at transistor level, but using the own logic resources available in the FPGA. Moreover, the design based on FPGAs implies the use of software design tools provided by FPGA or third-party vendors. The CP identification process, as well as the design, placement, and implementation of the aging sensors must be carried out using such tools.

Therefore, in the context of FPGA design, two significant constraints exist: 1) the available, built-in hardware resources; and 2) the design tools. What is proposed in this paper is to use standard embedded FPGA resources and vendor tools to build and place aging sensors wherever necessary for any circuit configured in the FPGA.

Application-dependent aging sensors for FPGAs proposed in the literature [20]–[22] are based on the detection of the occurrence of signal transitions in the CPs within a given observation interval T_g close to the active clock edge. The solution proposed in [21] requires a very complex sensor setup procedure, which prevents its widespread use, except for very simple circuits. It requires the delays in several paths of the circuit to be compensated by a very specific placement of the resources that implement each sensor. This causes the observation time interval to be nearly impossible to be accurately determined. Therefore, loosely defined T_g negatively impacts sensor's sensitivity.

In [20], authors presented the architecture and behavior of a novel aging sensor topology for reconfigurable circuits, as well as proof-of-concept tests conducted on simple circuits, where only one sensor was required. This sensor was manually inserted

in the design. The solution proposed in [22] is similar to the one in [20]. The main difference is that it allows the duration of the observation interval to be adjusted online, taking advantage of the variable phase-shift feature of Xilinx' digital clock managers (DCMs). No actual aging experiments were reported in these previous works. In this paper, we propose a methodology for FPGA design with embedded aging monitoring, using standard FPGA resources and vendor design tools, requiring limited human interaction, and allowing automatic multisensor insertion. Xilinx devices and design tools are used as test vehicles. The methodology is demonstrated in complex combinational and sequential circuits. The sensor's correct functionality and performance are also ascertained (both before and after aging of the devices), and low area and power overhead computed.

The remainder of this paper is organized as follows. In Section II, the topology of the aging sensor is presented. The proposed design methodology, which can be easily integrated in a standard FPGA design flow, is described in Section III. In Section IV, experimental results (including aging experiments based on the standard US MIL-STD-883 Method 1015.50 "Burn-In Test") are presented for several test circuits of increasing complexity. Finally, Section V summarizes the main conclusions and provides guidelines for future work.

II. PROPOSED AGING SENSOR TOPOLOGY

The principle of operation of the proposed sensor is based on the definition of an observation (or guardband) interval (T_g) in each clock period, close to the active clock edge, whose duration is programmable with a resolution (that depends on the operating frequency) in the range of tens of picoseconds. The sensor checks for the occurrence within the observation interval of transitions of the target signals (circuit's CPs, whose signal integrity is more likely to be compromised by aging effects). In normal operation, the signal is fast enough, so transitions do not take place during T_g . As the signal becomes slower due to aging effects (or due to other long-term effects, causing abnormal delays), the probability of a transition occurring within T_g increases. Whenever such a transition takes place, the sensor flags it, indicating that transitions are occurring too close to the active clock edge, and may thus cause signal integrity loss.

This technique has been referred to as circuit failure prediction [16], since the sensors can flag large delays before they cause system errors. The aforementioned resolution of the sensor is suitable enough for this purpose, because generous time-slack margins used in most designs (and in FPGA vendors' tools) prevent transitions from taking place some hundreds of picoseconds before the active clock edge.

Which signal paths should be monitored? Design tools allow the estimation of paths propagation delays, usually in a very conservative way, as simulation assumes pessimistic delays for the design corner—slow process, lower power supply voltage, and higher temperature. This allows the identification of critical paths (CP, the slowest combinational path between registers) or near CP. NBTI-dependent aging depends on temperature, electric field, and PMOS workload. Dependent on the application, on the synthesized configuration and on input vector

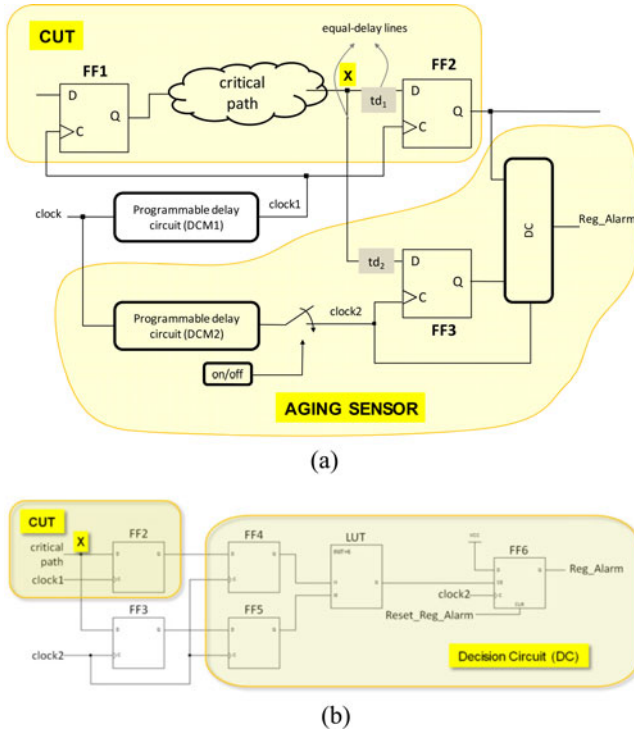


Fig. 1. (a) Block diagram of the proposed aging sensor (DC: decision circuit). (b) Details of the DC.

application, the inner tissue of the FPGA device will age differently in space and time. If FPGAs transistors' structure is known, it is possible to predict the areas that are more prone to be affected by aging effects, based on circuit functionality and gate workload probability. However, there are no FPGA tools currently available to identify the signals whose integrity is more likely to be compromised by aging effects. Therefore, in this study, it is assumed that those signals correspond to any path whose delay in nominal operating conditions exceeds $\alpha\%$ of the delay of the CP. Most experiments have been arbitrarily run with $\alpha = 90\%$.

The proposed sensor (see Fig. 1) has been implemented in a nanometer-scale Spartan-3AN FPGA device from Xilinx. As explained later, it uses only standard FPGA resources; therefore, it can also be implemented in other devices families from the same vendor (and, with minor modifications, in devices from other vendors).

In Fig. 1(a), the CP of the target circuit terminates at the input of flip-flop FF2. Two DCMs generate the circuit's main clock signal (clock1) and an auxiliary clock signal (clock2) with the same frequency, but exhibiting a negative phase shift equal to T_g , with regard to clock1. Thus, clock2 allows an early capture of the CP signal, T_g ahead of the capture performed by FF2 of the circuit under test (CUT). The system clock, clock1, is in fact delayed T_g with regard to clock2. For a given clock period T_{clk} , DCMs allow T_g to be configured to take any discrete value $T_g = N \times T_{clk}/256$, where N is an integer in the range $-255 < N < 255$. DCMs are standard resources available in current FPGAs [23], which, among other functionalities, allow such negative phase shifts to be applied to clock signals.

It is clear from Fig. 1 that FF3 captures the data in the CP a time T_g before FF2 does. Therefore, if the outputs of FF2 and FF3 differ, it follows that a transition of the CP signal, at node X, took place during T_g . In order to check for this situation, the decision circuit (DC) captures the outputs of FF2 and FF3 in another two flip flops (FF4 and FF5, respectively) that are activated by the rising edge of clock2. Reg_Alarm output continuously flags the occurrence of a transition of the CP signal during T_g , until FF6 is reset by a dedicated signal. The lookup table (LUT) in Fig. 1(b) implements an XOR function, whose output equals "1" at least during one clock cycle upon the occurrence of a transition during T_g .

Two improvements have been included in this topology with regard to the one first presented in [20]. In Fig. 1(a), it can be seen that the two DCMs share the same input signal, whereas in the former design the input of DCM2 was connected to the output of DCM1. This modification has been implemented because, according to the FPGA vendor, chain connection of DCMs may cause problems related to jitter. On the other hand, the active clock edge of FF4 and FF5 in Fig. 1(b) has been changed from the falling edge to the rising one. This allows the operation range of the sensor to be extended. In the sensor presented in [20], only $T_g < T_{clk}/2$ intervals can be observed. In this new version, the sensor can operate in the full T_g range.

As long as a given circuit is powered up, aging effects degrade both areas with low activity (or inactive) and with high activity, depending on technology, on circuit topology, functionality, and operation. However, in general, static NBTI stress leads to larger degradation than dynamic one, as dynamic operation allows the NBTI recovery process [24]. Aging sensors should be as robust as possible to aging effects, which would impact their timing performance. The proposed sensor fulfills this condition, as discussed in Section IV.

Unlike ASICs, in FPGAs, it is not possible to apply techniques (such as power gating) to disconnect sensors from the power supply voltage rail. Taking this into account, the proposed sensor includes an ON/OFF signal, which allows designers to decide whether the sensor will be operating continuously (to increase delay-fault capability) or disconnected from signal data path, except during short time intervals where sensors evaluate aging effects in the circuit (to reduce power consumption). This might also serve to mitigate aging effects in the sensor (by eliminating stress conditions) but, as stated previously, the proposed sensor is robust to these effects.

In order for the proposed sensor to work properly, the delays from node X (the output node of the CP) to the inputs of FF2 and FF3 must be the same, which means that $td_1 = td_2$. This condition can be accomplished in FPGAs, by placing both FF2 and FF3 in slices of the same configurable logic block (CLB, the basic logic building block of FPGAs [23]). This imposes a limitation to the placement of FF3, usually easy to comply with. Note that all remaining elements of the sensor can be freely placed in any available area of the device.

The sensor requires only very few (and standard) FPGA resources: one DCM, one LUT, and four flip flops (DCM and FF6 flip flop can be shared among different sensors). This implies low overhead per sensor, which is crucial as typical CUTs

require the monitoring of several CPs (or near CPs). Therefore, multisensor insertion is cost effective. The main limitation is the number of DCMs available in the FPGA device. However, DCM can be shared among different sensors. In fact, if all sensors require the same T_g interval (which is the common procedure), one DCM can be shared by all aging sensors. In this case, even FF6 can be shared, if a global Reg_Alarm signal is generated. Given the way target paths have been identified in this study, all sensors in the circuits analyzed in Section IV share the same DCM. In some cases, sensors are located in areas far away from each other in the FPGA fabric. Very interestingly, no skew problems affected any of the sensors in these cases. If a different approach is used to select the target paths, as for instance the one proposed in [21], different observation intervals may be required by different sets of sensors. In this case, separate DCMs must be used for each sensor set. This is not a severe limitation in practice, because even the relatively simple FPGA device used in this study includes eight DCMs. Given its very low complexity, the impact of the sensor in terms of area overhead and power consumption is not significant, as demonstrated by the experimental results presented in Section IV.

Since the sensor is located at the output of CPs, the loading effect on the critical delay needs to be carefully evaluated. This effect is circuit dependent, mainly because of interconnects. In all configured circuits where the experimental results have been obtained, the sensor inclusion caused a small increase of the critical delay in the range 30–91 ps. Hence, performance overhead is limited.

One very important characteristic of the proposed sensor is that its behavior is insensitive to power supply variations in the FPGA core (to the limit that causes functional collapse of the device). The critical issue in this regard is the matching of the delays $td1$ and $td2$ [see Fig. 1(a)] which, as mentioned previously, is guaranteed by construction.

III. SENSOR INSERTION

The process of inserting the required aging sensors in a given FPGA design should be based on the use of readily available tools, in order to allow the proposed solution to be widely adopted by designers. As the experimental results presented in Section IV have been obtained with Xilinx devices, the implementation of the proposed sensor insertion methodology using ISE design suite is described later. The methodology has been proved to work with different versions of a vendor tool (namely ISE 11.1, ISE 12.4, and ISE 13.1 from Xilinx), but it can be easily adapted to other vendor or third-party tools.

The steps required to insert aging sensors in a target circuit are as follows.

- 1) The design is synthesized from a standard description, where aging effects are not considered, to obtain the corresponding place and route information (which defines the FPGA configuration).
- 2) Timing analysis tools (PlanAhead) are used to identify CPs, as well as those whose delays exceed 90% (the aforementioned arbitrary value of α chosen) of the critical one.

Aging sensors have to be placed at the end of all those paths (see node X in Fig. 1).

- 3) The operating frequency ($1/T_{clk}$) determines the resolution of the sensors ($T_{clk}/256$) and this, in turn, the limit values of the observation interval T_g . From the target values of T_g , the sensor blocks can be configured. These blocks must be placed and routed in the FPGA fabric without modifying the timing behavior of the original circuit; otherwise, CPs or critical delays may be modified. Using PlanAhead tool, placement information is stored in a user's constraints file (.ucf) to force placement to be unchanged in subsequent syntheses (freezing the placement).
- 4) Available slices are identified (if they exist) in the CLB where the flip flop terminating each target path (FF2 in Fig. 1) is implemented, to locate FF3 flip flops (see Fig. 1) on them. This can be forced by storing the corresponding locations in the constraints file. These slices should be of the same type (M or L) as the slices where FF2 flip flops are located. The inputs of FF2 flip flops must be available, which is not usually the case, because this implies that the logic feeding these flip flops is located in a separate slice. Therefore, FF2 flip flops should usually be moved to unused CLBs. This action has to be performed directly by the designer, looking for free slices using the FPGA Editor tool and storing in the constraint file that slice as the location of the target FF2 and the corresponding FF3 flip flops. This is the first of the only two extra actions that might be needed from the designer, which can obviously be easily done.
- 5) Sensors are added to the corresponding paths in the original description of the system. This new description is synthesized with the constraints stored in the .ucf file to obtain the final circuit. When using a hardware description language description as design entry, a target path may end in an intermediate variable generated during the first synthesis process. In such a case, this variable cannot be located in the original description of the system, which cannot therefore be modified to connect it to the input of the corresponding sensor. The solution is to add the corresponding sensor to the original description of the system with its data input connected to any free input pin. After synthesizing the new description, the variable and the input of the sensor can be located by the FPGA editor tool and connected. This is the second and last extra action that might be required from the designer. Again, it can be easily done.

It can be concluded that it is practically feasible to use standard FPGA design flow and design tools to insert the proposed aging sensors in any FPGA-based design flow.

IV. EXPERIMENTAL RESULTS

Experiments using an actual FPGA device (a nanometer-scale Spartan-3AN FPGA from Xilinx) have been carried out to characterize the relationship between the increase in the CP delays of a given circuit and the values of T_g that detect them. Such

characterization allows ascertaining the sensor's correct functionality and performance, as well as their low area and power overhead. The validity of the sensor has also been proved after exposing the FPGA to accelerated aging conditions, demonstrating its suitable behavior in aged devices.

A. Experiments Before Device Aging

Aging acceleration can be performed by exposing the FPGA to an elevated temperature and core V_{DD} voltage [25]. However, this is usually an irreversible process, degrading the device. Hence, before carrying out aging experiments, another nondestructive test approach was followed to emulate aging effects for an initial functional validation of the sensor. Basically, NBTI-induced aging increases the time response of PMOS. Abnormal delays in signal paths can be injected by means of several parametric effects, namely through core V_{DD} down grading. Although $|V_{thp}|$ degradation depends on the workload of individual PMOS transistors, in a CP there is cumulative delay in the chain of individual gates, causing an equivalent average degradation level. This makes reasonable, as an initial approximation, to emulate aging effects by lowering core V_{DD} .

As discussed in Section II, since $td1 = td2$ (see Fig. 1), the behavior of the sensor is not strongly influenced by power supply variations affecting the core of the FPGA. Therefore, advantage has been taken of this fact to emulate the occurrence of aging effects by decreasing the power supply voltage of the FPGA core. Propagation delay variation models previously proposed by the authors [26] have been used to relate the variations of the core power supply voltage ΔV_{DD} with the variations of the propagation delay ΔT_{pd} of the CP. These models have been demonstrated [20], [26] to allow users to accurately determine the core power supply voltage value at which a given circuit will fail because of desynchronization caused by increased delays.

The results of proof-of-concept tests (conducted on simple circuits) regarding sensor topology are described in [20]. In this paper, the methodology has been applied to circuits of increasing complexity to validate the proposed approach (both sensor topology and design methodology) in a more realistic and practical scenario. To simplify the experiments, but without loss of generality, when using several sensors in a CUT, their outputs are combined in an OR operation to obtain a single aging alarm indication.

In all experiments, the nominal value of the FPGA core V_{DD} is 1.2 V. DCMs are powered by a supply voltage of 3.3 V (the maximum nominal one; usual nominal value is 2.5 V, so 3.3 V can be considered a somehow stressing condition with regard to using 2.5 V). The CPs and the corresponding delays in nominal conditions are obtained from the vendor design tools, as described in Section III.

Test vectors were stored in a ROM and applied to the CUT's inputs in a cyclic way. In order to avoid the ROM and associated circuits to be influenced by core V_{DD} reduction, these are configured in another FPGA device. Hence, two boards are used in the testbed: one with the configured CUT, and the other with the test pattern generator. Both boards operate with the same system clock frequency. For the different test vehicles, the

modules with larger delays are ISCAS benchmark modules, for which the test patterns proposed by J. Hayes are used.

Experiments were initially conducted to evaluate the impact of sensor insertion on the timing behavior of a circuit. The ISCAS-85 c432 benchmark was used for this purpose. Following the methodology proposed in Section III, four target paths were identified after the initial synthesis. In this case, the circuit operates at 50 MHz, resulting in a timing resolution of sensors of $20 \text{ ns}/256 = 78.125 \text{ ps}$ for T_g .

The synthesized circuit was repeatedly tested under decreasing FPGA core V_{DD} from its nominal value (1.2 V) until a faulty behavior was detected. A first run was applied to the circuit obtained from the original synthesis, for which the fault occurs at $V_{DD} = 0.734 \text{ V}$ (61.2% of nominal V_{DD}). Afterward, placement data from PlanAhead were included in the constraints file, the circuit was resynthesized and tested again, resulting in faulty operation at $V_{DD} = 0.746 \text{ V}$ (62.2% of nominal V_{DD}). The small difference is due to changes in routing (that was not frozen), indicating that these changes did not significantly modify delays. Finally, the four aging sensors required were inserted according to the proposed methodology, yielding a fault at $V_{DD} = 0.782 \text{ V}$ (65.2% of nominal V_{DD}), again very close to the original result.

The inclusion of the sensors resulted in an increase of the number of flip flops from 43 to 56 (23%) and of the number of slices from 70 to 90 (22%). Although these figures may seem high, it has to be taken into account that the c432 only uses very little (less than 1%) part of the resources available in the FPGA. In terms of available FPGA resources additional, flip flops account for 0.11% and additional slices for 0.33%.

Power consumption can be roughly estimated by the design tool. In this case, the insertion of the sensor causes an increase of 25% (31.7 mW). It should be noted that most of it (20%) comes from the additional DCM, likely because it is powered at 3.3 V.

A second set of experiments was conducted using the c432 circuit to evaluate the behavior of the sensors working with different values of T_g . The corresponding results are shown in Fig. 2(a), where the high linearity of the sensors can be clearly noticed. It can be also seen that larger observation intervals (T_g) imply a higher sensitivity of the sensor, as the alarm is flagged at higher core V_{DD} values (i.e., for smaller delay increases, meaning a less aged circuit). Moreover, if the designer selects, e.g., $N = 20$ (leading to a theoretical observation interval of $T_g = 1562.5 \text{ ps}$), the embedded sensors can predict circuit failure (without circuit error) with a user's define safety margin.

Similar experiments were conducted on a circuit implementing a 64-state Moore finite-state machine (FSM) using Gray coding and 8-bits input/output data. As in the previous case, the CUT operates at 50 MHz. Regarding the influence of sensor insertion on the timing behavior, the original circuit failed at $V_{DD} = 0.718 \text{ V}$ (59.8% of nominal V_{DD}), whereas with frozen placement it failed at $V_{DD} = 0.715 \text{ V}$ (59.6% of nominal V_{DD}), and at $V_{DD} = 0.752 \text{ V}$ (62.7% of nominal V_{DD}) once the three required sensors were inserted. It can be noticed that in this case, after freezing the placement, the synthesis resulted in a slightly faster circuit. A highly linear behavior of the sensors can again be noticed in Fig. 2(b).

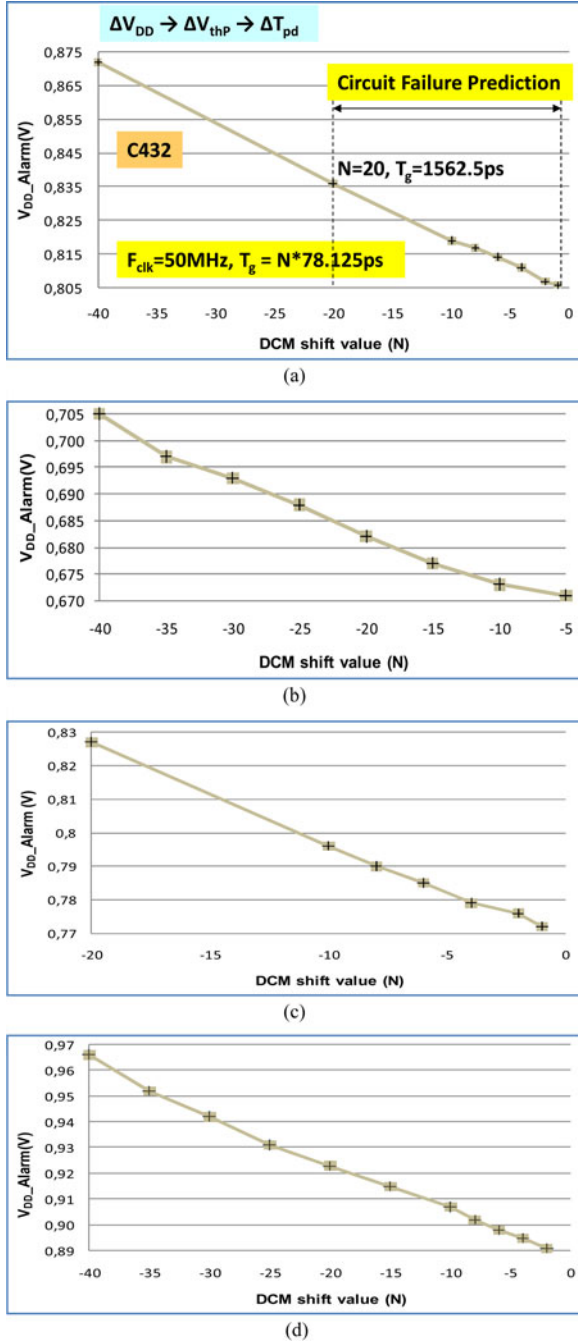


Fig. 2. Core V_{DD} at which the aging alarm is flagged in the emulation experiments as a function of the observation interval for (a) C432 circuit ($T_g = N \times 78.125ps$). (b) FSM circuit ($T_g = N \times 78.125ps$). (c) Arithmetic processor ($T_g = N \times 59.185ps$). (d) Circuit including the arithmetic processor and the c432 benchmark ($T_g = N \times 59.185ps$).

The inclusion of the sensors resulted in this case in an increase of the number of flip flops from 76 to 86 (11%) and of the number of slices from 68 to 69 (1.5%). In terms of available FPGA resources, these figures are 0.09% and 0.02%, respectively. In this case, the design tool estimates an increase in total power consumption of 33% (33.5 mW), again most of it (24.4%) due to the additional DCM.

The two circuits discussed previously require a very limited portion of the resources available in the FPGA. Hence, inserting sensors is quite easy according to the proposed methodology. The analysis of the results obtained from two much more complex circuits, shown later, demonstrates that the proposed solution behaves also very well when a significantly high portion of the resources are used by the target circuit. These two circuits operate at a higher frequency than the previous ones, 66 MHz, resulting in a different resolution for T_g (59.185 ps).

A complex arithmetic processor was implemented using 66% of FPGA slices. In it, the slowest module is a 16-bit multiplier (ISCAS c6288); thus, sensor insertion is performed in this module. Timing analysis indicated the need for seven sensors. Some of the target paths were associated to dedicated carry signals, which allowed the behavior of the sensors to be validated when connected not only to standard logic resources but also to specialized ones. The corresponding results are shown in Fig. 2(c).

The inclusion of the sensors resulted in an increase of the number of flip flops from 2238 to 2260 (1%) and of the number of slices from 3847 to 3886 (1%), showing the very low impact of sensor insertion in the case of circuits of reasonable practical size. In terms of available FPGA resources, additional flip flops account for 0.19% and additional slices for 0.67%. The estimated increase in total power consumption is 11.1% (36.1 mW) in this case, 8.2% due to the additional DCM.

Finally, the c432 benchmark was included in the circuit implementing the arithmetic processor, yielding 78% of slices used (a quite high figure, considering that because of the predefined set of resources within a FPGA, it is usually not possible to use most of them while complying with performance requirements). Timing analysis indicated four additional target paths, for a total of 11 sensors. The corresponding results are shown in Fig. 2(d).

In this case, the inclusion of the sensors resulted in an increase of the number of flip flops from 2383 to 2432 (2%) and of the number of slices from 4431 to 4479 (1%). In terms of available FPGA resources additional, flip flops account for 0.3% and additional slices for 0.6%.

The estimated increase in total power consumption is very similar to the previous cases (30 mW). Again, it mostly comes from the additional DCM (24.72 mW).

B. Aging Experiments

After the successful nondestructive tests discussed previously were completed, the FPGA was exposed to accelerated aging conditions (namely high temperature and core V_{DD} voltage above its nominal value of 1.2 V) while the CUT was in operation (i.e., switching activity existed due to the application of input stimuli). The experimental setup is shown in Fig. 3. The two FPGA boards were located in an oven with controlled temperature (80 °C during the experiments). An external oscillator was connected to the board under test to generate a stable global clock signal (the frequency of the oscillator included in the FPGA board is highly dependent on temperature, so it cannot be used for this kind of experiments, where aging must only affect the FPGA). A variable power supply source was used to set the core power supply voltage of the FPGA (V_{DD}) at 2.5 V



Fig. 3. Laboratory setup for the aging experiments.

(more than twice its nominal value). A frequency meter was connected to the output of one DCM to verify to what extent variations in temperature and power supply, as well as the aging process itself affect the operation of DCMs. Finally, the outputs of the CUT were connected to a logic analyzer.

It should be noted that there are standard procedures for aging experiments based on burn-in test with different goals.

- 1) "Screening to eliminate defective or marginal devices" described by US MIL-STD-883 Method 1015.50 "Burn-In Test". Burn-in screening is done usually at 125 °C during 168 h (one week), although there are different combinations of temperatures and durations proposed on the literature.
- 2) "Aging of the IC to estimate its mean steady life." The procedure is described by the automotive standard AECQ-100, which refers to JEDEC Standard No. 22, method A108D "Temperature, Bias, and Operating Life" and by MIL-STD 883 method 1005.9 "Steady-State Life." In both standards, different test conditions are described, recommending for digital ICs the high temperature operating life (HTOL test). The standard test condition is 1000 h (more than 41 days) at 125 °C, although an accelerated high temperature condition is also described by the MIL-STD.

In this paper, the method described in the standard US MIL-STD-883 Method 1015.50 "Burn-In Test" has been adapted to allow characterization of the aging of the devices over a certain period of time, which is our actual goal. Aging conditions have been adjusted to cause an aging as fast as possible, while avoiding the devices to be damaged. By doing so, we were able to measure the sensor performance as a function of aging for long periods of time (as in field operation) in the target nanometer technology.

The CUT used was the most complex one (arithmetic processor and c432 benchmark) from previous experiments, working at 66 MHz. As stated previously, this circuit requires most of the FPGA resources to be used. Three different values of T_g were tested after FPGA aging in order to verify if the linear behavior of the sensor in the nondestructive experiments is preserved after aging.

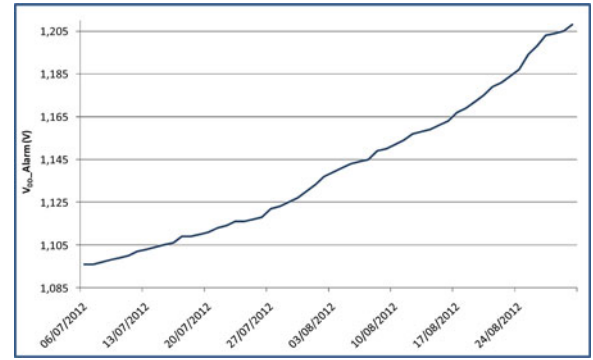


Fig. 4. Core V_{DD} at which the aging alarm is flagged as a function of time ($T_g = 3.78$ ns).

A first important conclusion is that, after three months of aging, there is no significant difference in the behavior of DCMs working at room temperature and supplied at 2.5 V and working at 80 °C, 3.3 V. No significant frequency variations of the clock output signal of the DCM were detected during the aging process. Hence, the frequency stability of the DCM was proved. This is a fundamental conclusion to validate the correct behavior of the aging sensor because its efficiency depends on the stability of the observation interval T_g and therefore on the stability of the DCMs.

In order to characterize the FPGA aging process, the delays of the CUT have been indirectly measured at different times by measuring the core power supply voltages at which 1) the sensor output is activated; and 2) the circuit fails because of desynchronization. To perform these measurements, the operating conditions are first restored to room temperature and nominal core V_{DD} , and then voltage is gradually decreased until conditions 1 and 2 occur. According to the standard, measurements are always performed within the first hour (actually within the first 5 min) after accelerated aging is stopped, to avoid annealing effects that could mask delay measurements on the CUT (and also to allow repeatability of the test in possible future sessions). After they are completed, aging conditions are resumed and the aging process goes on. As expected, the V_{DD} voltages at which 1 and 2 occur are higher as the circuit ages, indicating that delays are increasing because of this effect.

Fig. 4 shows some of the measured results demonstrating the aging process. As the DCM does not vary significantly from nominal conditions to high temperature and maximum operating voltage conditions and since the timing behavior of the rest of the sensor is compensated by design (as discussed in Section II and Fig. 1, $td1 = td2$), what was detected with the experiments is nearly exclusively the effect of the aging of the CUT.

Once the FPGA was aged for three months, a new experiment was performed to validate the linearity of the sensor, using the same highly complex CUT. The FPGA was configured with three different values of T_g , namely 3.78 ns (25% of T_{clk}), 3.31 ns (21.8%), and 1.93 ns (12.5%) and the V_{DD} voltages at which 1) the sensor output is activated; and 2) the CUT fails were again measured. Results of 1 are shown in Fig. 5. Regarding the core voltage at which CUT fails, the results were 1.015, 1.017,

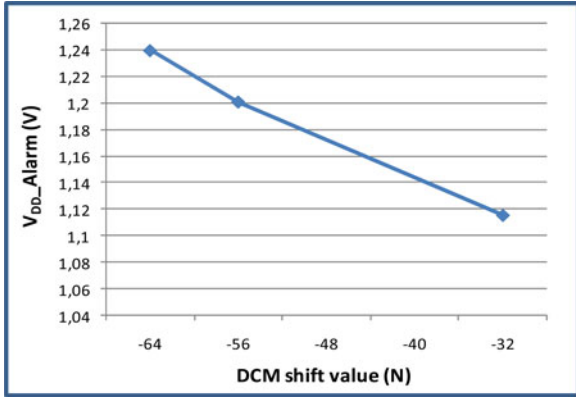


Fig. 5. Core V_{DD} at which the aging alarm is flagged as a function of the observation interval ($T_g = N \times 59.185$ ps).

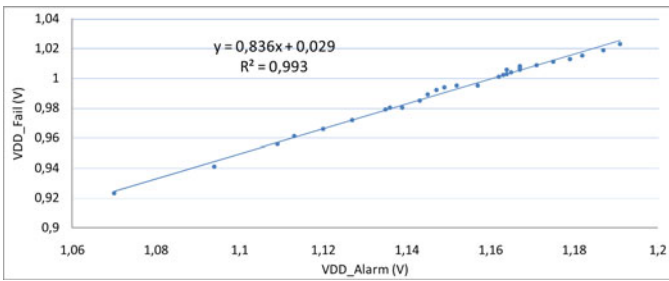


Fig. 6. Core V_{DD} at which the aging alarm is flagged as a function of the core V_{DD} at which circuit fails.

and 1.013 V for each value of T_g tested. These results assure that the voltage at which the CUT fails slightly differs when the sensor is programmed with different values of T_g .

Finally, although the stability of the DCMs was previously proved, it is important to demonstrate that the observation interval of the sensor, T_g , is not affected by any other factor, as the logic of the sensor itself, causing nonnegligible variations over time.

One way to check this is by analyzing the relationship between the core power supply voltages at which the sensor output is activated (V_{DD_Alarm}) and the one at which the circuit fails because of desynchronization (V_{DD_Fail}). Fig. 6 shows such a relation. A linear regression of measured data demonstrates that the relationship between V_{DD_Alarm} and V_{DD_Fail} is practically linear. That is, both V_{DD_Alarm} and V_{DD_Fail} increase in the same way due to the aging process. Fig. 6 also shows the equations resulting of the linear regression. The value of $R^2 = 0.993$ is very close to the ideal value for a perfect regression matching (1.0).

To further emphasize the linear behavior of the observation interval, Fig. 7 shows the value of V_{DD_Fail}/V_{DD_Alarm} as a function of V_{DD_Alarm} . The mean V_{DD_Fail}/V_{DD_Alarm} value is over 0.86 V and the maximum variation is 0.003 V, which means that the slope of the curve in Fig. 6 is nearly constant (i.e., it is nearly a straight line).

The results of the aging experiments demonstrate the following.

- 1) Under aging conditions, the sensor adequately flags aging.

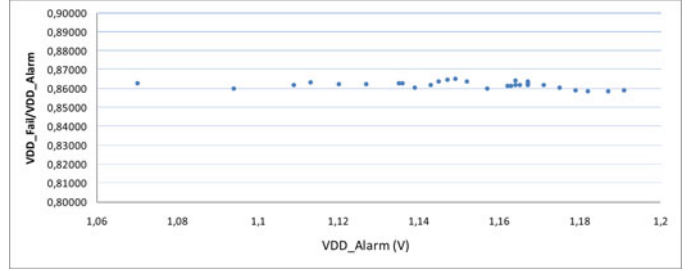


Fig. 7. V_{DD_Fail}/V_{DD_Alarm} value as a function of V_{DD_Alarm} .

- 2) DCMs are stable with temperature and voltage variations over time, so the observation interval of the sensor is also stable.
- 3) As a consequence of 2, aging of the sensor does not affect its functionality. Since the observation interval is stable, under aging conditions the sensor is correctly activated when the circuit is about to fail (with a predefined safety margin). The activation or not of the sensor output depends only on the degree of CUT aging.
- 4) The behavior of the CUT is not affected by the use of different T_g values.
- 5) The behavior of the sensor is highly linear and, under aging conditions, linearity is not significantly affected.

V. CONCLUSION

In this paper, a new aging sensor, as well as a methodology for FPGA design with embedded aging monitoring, has been presented. A very important feature is that only standard FPGA resources and design tools, with a very limited human interaction, are required. The methodology has been demonstrated in complex combinational and sequential circuits showing that multisensor insertion is practically feasible. The sensor's correct functionality and performance has been also ascertained. The impact of inserting several sensors in simple and complex circuits in terms of area and power overhead has been computed, demonstrating that the overhead is low enough not to degrade the performance of the circuits under test. Finally, the usefulness of the proposed approach has been validated in accelerated aging experiments.

The impact on the proposed methodology of scaling down the technology will not be relevant, as long as future FPGA devices use similar hardware resources as building blocks (which is the case for the new families of devices most recently released). Therefore, the strategy of embedding delay sensors to monitor aging using standard on-chip FPGA resources and design tools is expected to be rewardingly applicable to future node technologies.

As a future work, propagation delay variation models, previously proposed by the authors in [26], must be integrated in this methodology to precisely determine the required value of T_g in each case. The robustness of the sensor to temperature and process variations needs to be further investigated. Also, the effect of clock jitter on the stability of the observation interval T_g must be characterized. Finally, the possibility

of using technology-independent third-party tools for sensor insertion must be evaluated. As sensor description is technology independent, only sensor insertion procedure may need to be modified if third-party tools are used.

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